

Comprehensive Guide to Probability Distributions

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1 Introduction

This document provides a complete list of major probability distributions, organized by type (discrete vs. continuous). Each entry includes an explanation, real-world applications, expectation, variance, characteristic function, and a worked example with answer.

2 Discrete Distributions

2.1 Bernoulli Distribution

Explanation: A single trial with two outcomes: success (1) or failure (0).

Applications: Coin flip, pass/fail test, click/no-click in advertising.

$$\text{Expectation} = p$$

$$\text{Variance} = p(1 - p)$$

$$\text{Characteristic Function} = 1 - p + pe^{it}$$

Example: A biased coin has probability $p = 0.7$ of landing heads (success). What is the probability of success? Find the mean and variance.

Answer:

$$P(\text{success}) = 0.7$$

$$E[X] = 0.7$$

$$\text{Var}(X) = 0.7(1 - 0.7) = 0.21$$

2.2 Binomial Distribution

Explanation: Number of successes in n independent Bernoulli trials.

Applications: Number of defective items in a batch, number of heads in 10 coin flips.

$$\text{Expectation} = np$$

$$\text{Variance} = np(1 - p)$$

$$\text{Characteristic Function} = (1 - p + pe^{it})^n$$

Example: A fair coin is flipped 10 times. Find the probability of exactly 6 heads, the expected number of heads, and the variance.

Answer:

$$P(X = 6) = \binom{10}{6} (0.5)^6 (0.5)^4 = 210 \times 0.015625 \times 0.0625 = 0.2051$$

$$E[X] = 10 \times 0.5 = 5$$

$$\text{Var}(X) = 10 \times 0.5 \times 0.5 = 2.5$$

2.3 Geometric Distribution

Explanation: Number of trials until the *first* success.

Applications: Number of job interviews until hired, number of light bulbs tested until one fails.

$$\begin{aligned}\text{Expectation} &= \frac{1}{p} \\ \text{Variance} &= \frac{1-p}{p^2} \\ \text{Characteristic Function} &= \frac{pe^{it}}{1-(1-p)e^{it}}\end{aligned}$$

Example: The probability of winning a game is $p = 0.2$. What is the expected number of games until the first win? Find the variance.

Answer:

$$\begin{aligned}E[X] &= \frac{1}{0.2} = 5 \text{ games} \\ \text{Var}(X) &= \frac{1-0.2}{(0.2)^2} = \frac{0.8}{0.04} = 20\end{aligned}$$

2.4 Negative Binomial (Pascal) Distribution

Explanation: Number of trials until r successes occur.

Applications: Number of free throws attempted to make 5 shots, number of patients to see to get 3 recoveries.

$$\begin{aligned}\text{Expectation} &= \frac{r}{p} \\ \text{Variance} &= \frac{r(1-p)}{p^2} \\ \text{Characteristic Function} &= \left(\frac{pe^{it}}{1-(1-p)e^{it}} \right)^r\end{aligned}$$

Example: A basketball player has success probability $p = 0.6$. What is the expected number of attempts to make 3 shots? Find the variance.

Answer:

$$\begin{aligned}E[X] &= \frac{3}{0.6} = 5 \text{ attempts} \\ \text{Var}(X) &= \frac{3(1-0.6)}{(0.6)^2} = \frac{3 \times 0.4}{0.36} = \frac{1.2}{0.36} \approx 3.33\end{aligned}$$

2.5 Poisson Distribution

Explanation: Number of events in a fixed interval of time/space, given constant average rate λ and independence.

Applications: Number of emails per hour, car accidents at an intersection per day, radioactive decays.

$$\text{Expectation} = \lambda$$

$$\text{Variance} = \lambda$$

$$\text{Characteristic Function} = e^{\lambda(e^{it}-1)}$$

Example: A call center receives an average of 5 calls per minute ($\lambda = 5$). Find the probability of exactly 3 calls in a minute, the expected number, and variance.

Answer:

$$P(X = 3) = \frac{e^{-5} \cdot 5^3}{3!} = \frac{0.006738 \times 125}{6} = 0.1404$$

$$E[X] = 5$$

$$\text{Var}(X) = 5$$

2.6 Hypergeometric Distribution

Explanation: Number of successes in n draws *without replacement* from a finite population.

Applications: Probability of drawing 2 aces in a poker hand, quality control without replacement.

$$\text{Expectation} = n \frac{K}{N}$$

$$\text{Variance} = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$$

$$\text{Characteristic Function} = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}} \quad (\text{no simple closed form})$$

Example: A box contains 10 items: 4 defective ($K=4$) and 6 good. You draw 3 items without replacement ($n=3$). What is the probability of exactly 2 defective items? Find the mean.

Answer:

$$P(X = 2) = \frac{\binom{4}{2} \binom{6}{1}}{\binom{10}{3}} = \frac{6 \times 6}{120} = \frac{36}{120} = 0.3$$

$$E[X] = 3 \times \frac{4}{10} = 1.2$$

2.7 Discrete Uniform Distribution

Explanation: Finite set of equally likely integer outcomes.

Applications: Rolling a fair die, random number generation.

$$\text{Expectation} = \frac{a+b}{2}$$

$$\text{Variance} = \frac{(b-a+1)^2 - 1}{12}$$

$$\text{Characteristic Function} = \frac{e^{ita} - e^{it(b+1)}}{(b-a+1)(1-e^{it})}$$

Example: Roll a fair six-sided die (a=1, b=6). Find the expected value, variance, and probability of rolling a 4.

Answer:

$$E[X] = \frac{1+6}{2} = 3.5$$

$$Var(X) = \frac{(6-1+1)^2 - 1}{12} = \frac{6^2 - 1}{12} = \frac{35}{12} \approx 2.917$$

$$P(X = 4) = \frac{1}{6} \approx 0.1667$$

3 Continuous Distributions

3.1 Uniform (Rectangular) Distribution

Explanation: Constant probability density over $[a, b]$.

Applications: Random arrival time, quantization error in rounding.

$$\begin{aligned}\text{Expectation} &= \frac{a+b}{2} \\ \text{Variance} &= \frac{(b-a)^2}{12} \\ \text{Characteristic Function} &= \frac{e^{itb} - e^{ita}}{it(b-a)}\end{aligned}$$

Example: A bus arrives uniformly between 0 and 10 minutes. Find the probability of waiting less than 3 minutes, the mean wait time, and variance.

Answer:

$$\begin{aligned}P(X < 3) &= \frac{3-0}{10-0} = 0.3 \\ E[X] &= \frac{0+10}{2} = 5 \text{ minutes} \\ \text{Var}(X) &= \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.33\end{aligned}$$

3.2 Normal (Gaussian) Distribution

Explanation: Symmetric, bell-shaped curve; central limit theorem basis.

Applications: Heights, test scores, measurement errors.

$$\begin{aligned}\text{Expectation} &= \mu \\ \text{Variance} &= \sigma^2 \\ \text{Characteristic Function} &= e^{i\mu t - \sigma^2 t^2/2}\end{aligned}$$

Example: IQ scores are normally distributed with $\mu = 100, \sigma = 15$. Find the probability that a random person has IQ ≥ 130 , and the 90th percentile.

Answer:

$$P(X > 130) = P\left(Z > \frac{130-100}{15}\right) = P(Z > 2) = 0.0228 \text{ (2.28\%)}$$

$$90\text{th percentile: } z_{0.90} = 1.2816 \Rightarrow x = 100 + 1.2816 \times 15 = 119.22$$

3.3 Exponential Distribution

Explanation: Time until next event in a Poisson process (memoryless).

Applications: Waiting time for next customer, lifetime of electronic components.

$$\text{Expectation} = \frac{1}{\lambda}$$

$$\text{Variance} = \frac{1}{\lambda^2}$$

$$\text{Characteristic Function} = \left(1 - \frac{it}{\lambda}\right)^{-1}$$

Example: The time between customer arrivals has $\lambda = 0.5$ per minute. Find the probability of waiting more than 3 minutes, the mean wait time, and variance.

Answer:

$$P(X > 3) = e^{-0.5 \times 3} = e^{-1.5} = 0.2231$$

$$E[X] = \frac{1}{0.5} = 2 \text{ minutes}$$

$$\text{Var}(X) = \left(\frac{1}{0.5}\right)^2 = 4$$

3.4 Gamma Distribution

Explanation: Generalization of exponential; time until k events in Poisson process.

Applications: Rainfall totals, insurance claim sizes.

$$\text{Expectation} = \frac{k}{\theta}$$

$$\text{Variance} = \frac{k}{\theta^2}$$

$$\text{Characteristic Function} = (1 - i\theta t)^{-k}$$

Example: For a Gamma distribution with shape $k = 2$ and rate $\theta = 3$, find the mean and variance.

Answer:

$$E[X] = \frac{2}{3} \approx 0.667$$

$$\text{Var}(X) = \frac{2}{9} \approx 0.222$$

3.5 Chi-Squared Distribution

Explanation: Sum of squares of k independent standard normals.

Applications: Goodness-of-fit test, variance estimation.

$$\text{Expectation} = k$$

$$\text{Variance} = 2k$$

$$\text{Characteristic Function} = (1 - 2it)^{-k/2}$$

Example: For a chi-squared distribution with $k = 5$ degrees of freedom, find the mean, variance, and the probability of exceeding 11.07.

Answer:

$$\begin{aligned}E[X] &= 5 \\Var(X) &= 10 \\P(X > 11.07) &= 0.05 \text{ (critical value from table)}\end{aligned}$$

3.6 Beta Distribution

Explanation: Distribution on $[0, 1]$ with two shape parameters α, β .

Applications: Bayesian conjugate prior for binomial proportion, PERT project management.

$$\begin{aligned}\text{Expectation} &= \frac{\alpha}{\alpha + \beta} \\ \text{Variance} &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} \\ \text{Characteristic Function} &= {}_1F_1(\alpha; \alpha + \beta; it) \text{ (Confluent hypergeometric)}\end{aligned}$$

Example: For a Beta distribution with $\alpha = 2, \beta = 5$, find the mean and variance.

Answer:

$$\begin{aligned}E[X] &= \frac{2}{2 + 5} = \frac{2}{7} \approx 0.2857 \\ Var(X) &= \frac{2 \times 5}{(7)^2(8)} = \frac{10}{49 \times 8} = \frac{10}{392} \approx 0.0255\end{aligned}$$

3.7 Log-Normal Distribution

Explanation: $\ln(X)$ is normal; positive-only, right-skewed.

Applications: Stock prices, income distributions.

$$\begin{aligned}\text{Expectation} &= e^{\mu + \sigma^2/2} \\ \text{Variance} &= (e^{\sigma^2} - 1)e^{2\mu + \sigma^2} \\ \text{Characteristic Function} &= \text{No simple closed form}\end{aligned}$$

Example: If $\ln(X) \sim N(\mu = 1, \sigma^2 = 0.25)$, find $E[X]$ and $Var(X)$.

Answer:

$$\begin{aligned}E[X] &= e^{1+0.25/2} = e^{1.125} \approx 3.077 \\ Var(X) &= (e^{0.25} - 1)e^{2 \times 1 + 0.25} = (0.2840)e^{2.25} = 0.2840 \times 9.4877 \approx 2.695\end{aligned}$$

3.8 Weibull Distribution

Explanation: Flexible lifetime distribution; can model increasing/decreasing hazard rates.

Applications: Wind speed, reliability engineering.

$$\text{Expectation} = \lambda \Gamma \left(1 + \frac{1}{k} \right)$$

$$\text{Variance} = \lambda^2 \left[\Gamma \left(1 + \frac{2}{k} \right) - \Gamma^2 \left(1 + \frac{1}{k} \right) \right]$$

Characteristic Function = No simple closed form

Example: For Weibull with scale $\lambda = 2$ and shape $k = 1.5$, find $E[X]$ (use $\Gamma(1.6667) = 0.9027$, $\Gamma(2.3333) = 1.2636$).

Answer:

$$E[X] = 2 \times \Gamma(1 + 1/1.5) = 2 \times \Gamma(1.6667) = 2 \times 0.9027 = 1.8054$$

$$\text{Var}(X) = 4[1.2636 - (0.9027)^2] = 4[1.2636 - 0.8149] = 4 \times 0.4487 = 1.7948$$

3.9 Cauchy Distribution

Explanation: Heavy tails; no finite mean or variance (undefined).

Applications: Resonant behavior in physics, ratio of normals.

Expectation = Undefined

Variance = Infinite (undefined)

$$\text{Characteristic Function} = e^{i\mu t - \gamma|t|}$$

Example: For the standard Cauchy distribution (median = 0), find the median and interquartile range.

Answer:

$$\text{Median} = 0$$

$$\text{IQR} = 2 \text{ (Q1} = -1, \text{ Q3} = 1)$$

3.10 Laplace (Double Exponential) Distribution

Explanation: Sharper peak than normal, heavier tails.

Applications: Laplace noise in differential privacy, financial returns.

$$\text{Expectation} = \mu$$

$$\text{Variance} = 2b^2$$

$$\text{Characteristic Function} = \frac{e^{i\mu t}}{1 + b^2 t^2}$$

Example: Laplace distribution with $\mu = 0, b = 1$. Find $P(|X| < 1)$, mean, and variance.

Answer:

$$P(|X| < 1) = 1 - \frac{1}{2}e^{-1} - \frac{1}{2}e^{-1} = 1 - e^{-1} = 0.6321$$

$$E[X] = 0$$

$$\text{Var}(X) = 2 \times 1^2 = 2$$

3.11 Pareto Distribution

Explanation: Power-law tail; “80-20 rule.”

Applications: Wealth distribution, city populations.

$$\text{Expectation} = \frac{x_m \alpha}{\alpha - 1} \quad \text{for } \alpha > 1$$

$$\text{Variance} = \frac{x_m^2 \alpha}{(\alpha - 1)^2 (\alpha - 2)} \quad \text{for } \alpha > 2$$

$$\text{Characteristic Function} = \alpha (x_m i t)^{\alpha} \Gamma(-\alpha, -i x_m t) \quad (\text{complicated})$$

Example: Pareto with $x_m = 1, \alpha = 3$. Find $E[X]$ and $Var(X)$.

Answer:

$$E[X] = \frac{1 \times 3}{3 - 1} = \frac{3}{2} = 1.5$$

$$Var(X) = \frac{1^2 \times 3}{(2)^2 (1)} = \frac{3}{4} = 0.75$$

4 Quick Reference Table

Table 1: Summary of all distributions with expectation, variance, and characteristic function

Distribution	Type	Expectation	Variance	Characteristic Function
Bernoulli	Discrete	p	$p(1-p)$	$1-p+pe^{it}$
Binomial	Discrete	np	$np(1-p)$	$(1-p+pe^{it})^n$
Geometric	Discrete	$1/p$	$(1-p)/p^2$	$\frac{pe^{it}}{1-(1-p)e^{it}}$
Neg. Binomial	Discrete	r/p	$r(1-p)/p^2$	$\left(\frac{pe^{it}}{1-(1-p)e^{it}}\right)^r$
Poisson	Discrete	λ	λ	$e^{\lambda(e^{it}-1)}$
Hypergeometric	Discrete	nK/N	$n\frac{K}{N}(1-\frac{K}{N})\frac{N-n}{N-1}$	No simple closed form
Discrete Uniform	Discrete	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$	$\frac{e^{ita}-e^{it(b+1)}}{(b-a+1)(1-e^{it})}$
Uniform (cont.)	Continuous	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{itb}-e^{ita}}{it(b-a)}$
Normal	Continuous	μ	σ^2	$e^{i\mu t-\sigma^2 t^2/2}$
Exponential	Continuous	$1/\lambda$	$1/\lambda^2$	$(1-it/\lambda)^{-1}$
Gamma	Continuous	k/θ	k/θ^2	$(1-i\theta t)^{-k}$
Chi-Squared	Continuous	k	$2k$	$(1-2it)^{-k/2}$
Beta	Continuous	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	${}_1F_1(\alpha; \alpha+\beta; it)$
Log-Normal	Continuous	$e^{\mu+\sigma^2/2}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$	No simple closed form
Weibull	Continuous	$\lambda\Gamma(1+1/k)$	$\lambda^2[\Gamma(1+2/k)-\Gamma^2(1+1/k)]$	No simple closed form
Cauchy	Continuous	Undefined	Infinite	$e^{i\mu t-\gamma t }$
Laplace	Continuous	μ	$2b^2$	$\frac{e^{i\mu t}}{1+b^2 t^2}$
Pareto	Continuous	$\frac{x_m\alpha}{\alpha-1}$	$\frac{x_m^2\alpha}{(\alpha-1)^2(\alpha-2)}$	$\alpha(x_m it)^\alpha \Gamma(-\alpha, -i\alpha x_m it)$